

Rajdeep Gill

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EDUCATION

University of Manitoba

Bachelor of Science, Computer Engineering (GPA: 4.45/4.5)

Winnipeg, MB

May 2026

WORK EXPERIENCE

Robinhood

Software Engineer Intern

May 2025 - Present

Toronto, ON

- Reduced customer margin calls by 37% and boosted margin trading activity by 23% by launching a real-time maintenance dashboard with embedded trade execution, surfacing key risk indicators like maintenance ratio and equity percentage
- Architected a cross-platform margin experience using a server-driven UI approach to unify business logic across Web, iOS, and Android, driving a 17% increase in margin usage
- Accelerated the adoption of a new Protobuf design system by implementing its foundational components, adding a logging framework to improve observability, and resolving initial implementation gaps
- Repaired the core integration suite for the margin experience and added 20+ tests, achieving 86% code coverage

Biosensors Research Lab

Undergraduate Research Assistant

May 2024 - Dec 2024

Winnipeg, MB

- Achieved 85% accuracy in predicting cell viability from 10,000+ diffraction images using a custom CNN, establishing a viable, non-invasive measurement method for cell health
- Built an interactive model validation GUI in Python that rendered SHAP values to provide real-time explanations of prediction drivers, enabling researchers to debug model behavior on a per-cell basis
- Architected an end-to-end data processing pipeline that boosted morphological parameter extraction accuracy by 17% and cut pre-processing runtime in half (50% reduction) using Python's multiprocessing library
- Replaced a high-risk system of manually-shared hard drives with a centralized MySQL database, eliminating data loss and versioning conflicts, and enabling complex, cross-experiment queries for the first time

Biosensors Research Lab

Undergraduate Research Assistant

May 2023 - Aug 2023

Winnipeg, MB

- Engineered a general-purpose Python control framework for the lab's oscilloscope, abstracting low-level hardware commands into a high-level API that empowered researchers to script and automate entire experimental sequences.
- Slashed data analysis runtime by over 90% (1 hour to <6 minutes) by refactoring legacy Python scripts, replacing inefficient loops with vectorized NumPy operations
- Developed a novel classification system using a Gaussian Mixture Model to determine cell viability from time-series impedance data, creating a probabilistic tool for assessing sample health

PROJECTS

Void - AI App Builder (Next.js, TypeScript, Prisma, Docker)

- Developed an app to generate and deploy web apps from user prompts, integrating a usage-based Stripe billing system to monetize user compute time and API token consumption
- Implemented a persistent background job queue to process long-running AI build tasks asynchronously, allowing users to submit jobs and receive the completed result at a later time, even after closing the browser
- Utilized Docker for user code sandboxing, secure code execution and real-time application previews

Collaborative Document Editor (React, TypeScript, Socket.io)

- Engineered a collaborative text editor for up to 20 simultaneous users by broadcasting text operations, cursor locations, and online statuses, via WebSockets, achieving a sub-250ms synchronization latency
- Integrated Gemini to provide real-time AI-powered suggestions for text, formatting and content generation

SKILLS

- Programming Languages:** Python, TypeScript, JavaScript, C/C++, HTML/CSS, Java
- Frameworks & Tools:** React, Next.js, PyTorch, Flask, Django, Express.js, AWS (EC2, S3), GCP, OAuth